Combining Technical Indicators and Metaheuristic Optimizers for Short-Term Neural Intraday Trend Prediction in Mini-Index Futures

Abstract

Short-term trend forecasting in high-frequency derivatives markets represents one of the most challenging applications of computational intelligence, characterized by severe non-stationarity, regime shifts, and low signal-to-noise ratios. Although supervised machine learning models and neural networks—such as Multilayer Perceptrons (MLPs), Support Vector Machines (SVMs), Random Forests (RFs), and 1D Convolutional Neural Networks (1D-CNNs)—are widely employed for trend classification, their generalization depends critically on input feature quality and hyperparameter optimization across non-convex search spaces. This article extends prior research by presenting a unified, controlled multi-optimizer benchmark framework for short-term financial time series prediction on 5-minute B3 Mini-Index futures (WIN) data from the BovDB repository. Using Information Gain feature selection to select a compact subset of 7 informative technical indicators from an initial pool of 66, five distinct optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—are evaluated under an identical computational budget cap of $N_{\text{eval}} = 1{,}500$ evaluations per seed. To prevent temporal look-ahead leakage, all models operate within a strict 60/20/20 sequential chronological split and a compound MCC-driven objective ($0.60 \times \text{MCC} + 0.40 \times F_1$). Experimental results demonstrate that evolutionary optimizers (GA and DE) coupled with MLP and 1D-CNN backbones achieve superior out-of-sample classification balance (MCC $= 0.231 \pm 0.012$ and 0.228 ± 0.014), significantly outperforming Random Search ($p < 0.01$, Cohen's $d > 1.2$). In out-of-sample financial backtesting with real-world transaction costs (0.01%), the GA-tuned neural classifier yields a Net Return of +18.4% and a Sharpe ratio of 1.42. These findings reinforce the value of combining feature selection with equal-budget metaheuristic optimization for quantitative trading.

Keywords: Metaheuristic benchmark · Neural network optimization · Particle swarm optimization · Genetic algorithm · Intraday trend classification · Systematic literature review · Brazilian futures market

ACC: Accuracy; AUC: Area Under the Curve; B3: Brasil Bolsa Balcão (Brazilian Stock Exchange); CNN: Convolutional Neural Network; DE: Differential Evolution; EMA:

Exponential Moving Average; GA: Genetic Algorithm; GWO: Grey Wolf Optimizer; MCC: Matthews Correlation Coefficient; MDD: Maximum Drawdown; ML: Machine Learning; MLP: Multilayer Perceptron; MOA: Metaheuristic Optimization Algorithm; PSO: Particle Swarm Optimization; RF: Random Forest; RS: Random Search; SMA: Simple Moving Average; SRL: Systematic Review of Literature; SVM: Support Vector Machine; WIN: Mini-Index Futures Contract.

1 Introduction

The financial market represents one of the most prominent arenas for allocating capital across diverse financial instruments, aiming to achieve economic returns while managing exposure to market risk [**? ?**]. Among the wide array of traded assets, derivative instruments—and specifically futures contracts—are particularly sought after by institutional and retail market participants due to their high liquidity, leverage potential, and operational utility for both risk hedging against price fluctuations and speculative trading strategies [1, 2]. These contracts establish an agreement to purchase or sell an underlying asset at a predetermined price on a future date, rendering them exceptionally sensitive to short-term market expectations, order flow dynamics, and macroeconomic announcements [3, 4].

In Brazil, Brasil Bolsa Balcão (B3) serves as the primary regulated exchange environment. Within B3's derivatives segment, the Mini-Index futures contract (symbol: WIN), referenced to the flagship Ibovespa stock index, exhibits immense daily trading volume and liquidity. WIN futures are heavily utilized in short-term quantitative trading strategies and high-frequency intraday execution, making their tick- and candle-level data a prime domain for computational intelligence studies in financial time series forecasting [5, 6]. However, financial price series derived from 5-minute intraday trading exhibit severe non-stationarity, low signal-to-noise ratios, structural regime shifts, and rapid dissipation of arbitrage patterns as market participants exploit temporary inefficiencies [7, 8].

Early attempts to formalize market patterns and predict directional movements relied on static linear models, such as autoregressive integrated moving average (ARIMA) techniques [**?**]. As finance and computer science converged under the computational intelligence paradigm [**?**], data-driven machine learning (ML) models gained widespread adoption. Supervised learning algorithms—including Random Forests (RF) [**?**], Support Vector Machines (SVM) [9], Multilayer Perceptrons (MLP) [10], and 1D Convolutional Neural Networks (1D-CNN) [11]—have demonstrated strong capabilities in learning complex, non-linear decision boundaries from historical market features.

A central prerequisite for building effective data-driven financial classifiers involves technical analysis, wherein mathematical transformations of price and volume series—such as Simple Moving Averages (SMA), Exponential Moving Averages (EMA), and standard deviation envelopes—are computed to capture trend momentum and volatility regimes [12**?**]. Nevertheless, computing high-dimensional pools of technical indicators introduces significant redundancy and noise, creating a high-dimensional feature space that degrades classifier generalization and increases computational overhead [13, 14]. Consequently, applying feature selection techniques to extract compact, highly informative indicator subsets becomes vital for stabilizing training dynamics [1, 15].

Beyond feature selection, classifier performance depends critically on hyperparameter tuning [16]. In neural network architectures such as MLPs and 1D-CNNs, structural parameters (e.g., hidden layer width, filter counts), regularization coefficients (L2 weight decay α, dropout rate p_{drop}), and optimization settings (learning rate lr, mini-batch size B) form a complex, non-convex global optimization surface over a mixed discrete-continuous search space $\boldsymbol{\theta} \in \mathbb{R}^D$ [8**?**]. Traditional gradient-based optimization algorithms (e.g., Stochastic Gradient Descent or Adam) often suffer from premature trapping in local sub-optima when traversing non-convex financial loss landscapes [17, 18]. To overcome these limitations, metaheuristic optimization algorithms (MOAs)—spanning evolutionary algorithms such as Genetic Algorithms (GA) [19] and Differential Evolution (DE) [20], as well as swarm intelligence paradigms such as Particle Swarm Optimization (PSO) [21] and Grey Wolf Optimizer (GWO) [17]—have gained substantial prominence in recent literature [2, 22].

Despite the growing literature on metaheuristic-tuned financial classifiers published in premier neural computing journals [2, 6, 8, 23], existing financial machine learning studies overwhelmingly suffer from three major methodological flaws:

1. **The Budget Inequality Flaw**: Benchmark studies frequently evaluate metaheuristics under unconstrained or unequal evaluation counts (e.g., comparing a GA running for 100 generations with a PSO running for 30 iterations) [8]. Because stochastic search performance scales monotonically with candidate evaluation count, reported performance superiority in prior studies is often an artifact of unequal evaluation budgets rather than inherent algorithmic efficiency.
2. **The Temporal Data Leakage Flaw**: Standard randomized K-fold cross-validation or randomized data shuffling is routinely applied to time series data [7, 24]. Shuffling 5-minute intraday bars breaks chronological continuity, allowing future price dynamics to contaminate training sets and producing artificially inflated validation metrics that fail under live trading execution.
3. **The Classification-Economic Disconnect Flaw**: Models are overwhelmingly tuned and evaluated solely on raw classification accuracy or cross-entropy loss [25]. However, raw accuracy treats false positive and false negative signals symmetrically and ignores class imbalance. In financial markets, uncalibrated signals incur severe transaction fees and slippage, leading to net losses despite high nominal accuracy.

To resolve these methodological limitations and extend our previous preliminary work [1, 24], this paper presents a controlled multi-optimizer benchmark framework for short-term neural intraday trend classification on 5-minute B3 Mini-Index futures data. Incorporating Information Gain feature selection to extract a compact 7-indicator subset from 66 raw technical indicators, we systematically evaluate five distinct optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer

(GWO)—across four diverse classifier backbones (MLP, SVM, RF, 1D-CNN). All optimizers operate under a strictly equal candidate evaluation budget cap of $N_{\text{eval}} = 1{,}500$ evaluations per seed within a leakage-free 60/20/20 chronological split and a compound MCC-driven objective ($0.60\text{MCC} + 0.40F_1$).

The principal technical contributions of this work are summarized as follows:

- **Formulate an equal-budget metaheuristic benchmark protocol** enforcing a strict budget cap of $N_{\text{eval}} = 1{,}500$ candidate evaluations per random seed across five distinct optimizer families operating over multi-model hyperparameter search spaces.
- **Implement a leakage-free sequential temporal validation protocol** partitioning 15,057 clean 5-minute B3 Mini-Index futures bars into a chronological 60% Training, 20% Validation, and 20% Out-of-Sample Test split (`shuffle = false`) to eliminate look-ahead data leakage.
- **Formulate a compound MCC-driven objective function** ($0.60\text{MCC}+0.40F_1$) designed to resist class imbalance distortion and maximize discriminative signal quality.
- **Demonstrate competitive parity among top-tier metaheuristics** through non-parametric Friedman omnibus tests ($\chi^2 = 38.42, p = 9.2 \times 10^{-8}$) and post-hoc Wilcoxon-Holm tests with Cohen's d effect sizes, proving that GA, DE, and PSO achieve statistically equivalent solution quality under budget equality.
- **Execute out-of-sample financial backtesting under transaction costs**, demonstrating that the GA-optimized neural classifier yields a Net Return of +18.4%, a Sharpe ratio of 1.42, and a Maximum Drawdown of −8.2% under standard exchange fees (0.01%).
- **Synthesize architectural design guidelines for financial AI deployment**, establishing that crossover recombination in evolutionary algorithms (GA and DE) provides superior resistance to local trapping compared to wolf leadership triad attraction (GWO).

The remainder of this manuscript is organized as follows: Section 2 presents a Systematic Review of Literature (SRL) on neural optimization and metaheuristics in financial forecasting. Section 3 details the materials and methods (dataset generation, Information Gain feature selection, search space formulations, metaheuristic operators, and ablation design). Section 4 presents empirical benchmark results, convergence dynamics, statistical hypothesis tests, and financial backtest evaluations. Section 5 discusses mechanistic search interpretations, scalability, and local trapping risks. Finally, Section 6 concludes the paper and outlines future research directions.

2 Systematic Review of Literature

With advancements in artificial intelligence and the widespread deployment of machine learning and deep learning methodologies across financial engineering [?], predicting future asset price movements has become a core area of computational finance research. This section presents a Systematic Review of Literature (SRL) examining recent research at the intersection of feature selection, metaheuristic optimization, and neural market forecasting, delineating key methodological paradigms and identifying open research gaps.

2.1 Machine Learning and Feature Selection in Market Forecasting

Early comparative surveys established the viability of supervised learning algorithms for market direction classification. Ballings et al. [**?**] conducted a large-scale empirical evaluation comparing Support Vector Machines, Random Forests, Kernel Factory, AdaBoost, and Logistic Regression on financial data from 5,767 European companies. Their findings identified SVM and Random Forest as the top-performing classifiers, demonstrating that predictive accuracy relies heavily on input feature quality, hyperparameter configuration, and model regularization.

Because technical analysis indicators introduce substantial redundancy when computed across multiple lookback windows, feature selection has emerged as a critical preprocessing step. Naik and Mohan [9] applied the Boruta feature selection algorithm to extract informative technical indicators from an initial set of 33 technical combinations, evaluating an ANN classifier on National Stock Exchange (NSE) data. Their feature selection protocol reduced prediction error by 12%, proving that filtering out uninformative indicators mitigates the curse of dimensionality and stabilizes neural network training. Similarly, Mutinda and Yong [11] proposed a sequential ensemble decision tree (SEDT) feature selection strategy combined with hybrid classification, demonstrating that feature space compacting improves classification sensitivity in volatile financial markets.

2.2 Metaheuristic Optimization of Financial Neural Networks

Hyperparameter optimization is vital for maximizing neural network generalization in non-stationary environments. Traditional gradient-based backpropagation algorithms (such as Stochastic Gradient Descent or Adam) often suffer from local entrapment when tuning network hyperparameter configurations over non-convex loss surfaces [8]. Ding et al. [16] introduced a hybrid Genetic Algorithm–Back-Propagation (GA-BP) framework, leveraging GA's global exploration capability to locate promising initial search regions before applying backpropagation fine-tuning.

In financial time series modeling, F. Ecer [10] investigated optimized MLP architectures for financial price forecasting, comparing multi-layer topologies and non-linear activation functions tuned via population-based metaheuristics. Rahim et al. [26] applied a single Genetic Algorithm to tune technical indicator combinations for 5-minute futures contracts, observing significant accuracy gains over untuned baselines. Bashir et al. [6] proposed EvoBagNet, an evolutionary bagging network incorporating a (1+1) evolutionary algorithm for equity price forecasting on the National Stock Exchange of India, demonstrating that evolutionary search operators enhance model robustness under non-stationary regimes.

More recently, Al Bannoud et al. [2] conducted a comprehensive benchmark evaluating 14 machine learning surrogate models coupled with eight metaheuristic optimization algorithms (including GWO, PSO, GA, and ABC) within a hybrid computational framework. Their work established that Grey Wolf Optimizer (GWO) and Particle Swarm Optimization (PSO) provide exceptional convergence efficiency and solution quality when optimizing complex neural surrogate parameters over non-differentiable objective landscapes.

2.3 Identification of Research Gaps and Synthesis

Despite significant progress reported across recent literature, a systematic analysis reveals critical methodological limitations in existing studies. Table 1 summarizes representative literature across five key dimensions: domain application, model family, optimization paradigm, validation protocol, and identified methodological gaps.

As detailed in Table 1, existing literature suffers from three pervasive limitations:

1. *Lack of Equal-Budget Benchmark Protocols*: Prior studies frequently compare metaheuristics under unequal candidate evaluation counts, making it impossible to determine whether reported performance gains stem from superior search operators or unconstrained computation [8].
2. *Widespread Temporal Data Leakage*: Studies routinely employ randomized K-fold cross-validation or data shuffling on financial time series, causing future market states to contaminate validation folds [7, 23].
3. *Disconnection from Real-World Trading Friction*: Classifiers are tuned exclusively on raw classification accuracy, ignoring class imbalance and live trading costs (brokerage fees and slippage) that deteriorate real-world profitability [1].

Unlike previous studies that treat feature selection, hyperparameter tuning, and economic evaluation separately, this work integrates Information Gain feature selection with an equal-budget ($N_{\text{eval}} = 1{,}500$) multi-optimizer benchmark (RS, GA, PSO, DE, GWO) across four model families (MLP, SVM, RF, 1D-CNN) operating under a strict 60/20/20 chronological split and an out-of-sample financial backtest with exchange transaction fees.

3 Materials and Methods

Figure 1 presents an overview of the computational workflow pipeline adopted in this study. The methodology spans six sequential stages: (i) acquisition of high-frequency 5-minute B3 Mini-Index futures data from BovDB, (ii) construction and min-max normalization of technical analysis indicators, (iii) Information Gain feature selection to extract a compact 7-indicator subset, (iv) formulation of multi-model hyperparameter search spaces across four classifier families (MLP, SVM, RF, 1D-CNN), (v) equal-budget metaheuristic optimization ($N_{\text{eval}} = 1{,}500$) under a leakage-free 60/20/20 sequential temporal split, and (vi) out-of-sample statistical, predictive, and financial evaluation.

Table 1: Systematic Literature Review (SRL) comparison matrix of representative studies in optimizer-driven financial forecasting and identified methodological gaps.

Study	Model & Domain	Optimization Paradigm	Validation Protocol	Identified Methodological Gaps
Ballings et al. (2015) [?]	Stock prediction, 5,767 European firms	Grid search / Manual	Cross-validation	Absence of metaheuristic hyperparameter search.
Ding et al. (2011) [16]	Neural networks, Benchmark data	Hybrid GA-BP	Standard split	No equal budget cap across baseline optimizers.
Naik & Mohan (2019) [9]	ANN trend, NSE Equities	Boruta + Default ANN	Train/Test split	Static hyperparameter tuning without metaheuristics.
Ecer et al. (2020) [10]	MLP classifier, Financial	GA and PSO tuning	K-fold CV	Evaluated single model under unconstrained budget.
Billah et al. (2024) [4]	Intraday trend, Equities	Moving average heuristics	Standard split	Absence of automated neural hyperparameter search.
Bareket & Pârv (2024) [22]	Market surge, Equities	Adaptive ML classifiers	Medium-term split	Evaluation restricted to static indicator sets.
Bandgar et al. (2025) [27]	Time series, Financial	Multi-model ensembles	Random split	Absence of candidate evaluation budget cap.
Rahim et al. (2025) [26]	Intraday futures, 5-min	Single GA optimization	Train/Test split	Evaluated single GA without multi-swarm benchmark.
Dhingra et al. (2025) [8]	Neural forecasting, Solar time series	Single metaheuristic	Sequential split	Unequal evaluation budget across baseline algorithms.
Mutinda & Yong (2026) [11]	Financial classification	SEDT Feature selection	K-fold CV	Feature selection without classifier budget tuning.
Simatupang et al. (2026) [23]	Stock index, IDX Equities	Bayesian Random Forest	Random K-fold CV	Randomized CV (Temporal data leakage risk).
Al Bannoud et al. (2026) [2]	ANN–ODE, Clinical kinetics	GWO, PSO, GA, ABC	Multi-partition CV	Non-financial domain; unconstrained evaluation budget.
Bashir et al. (2026) [6]	Stock price, NSE Equities	(1+1) Evolutionary Algo	Train/Test split	Evaluated single EA without multi-swarm benchmark.
Souza et al. (2025) [1]	RF classifier, B3 Futures	InfoGain + Default RF	Sequential split	Untuned baseline classifier without metaheuristics.
Souza et al. (2026) [24]	MLP/SVM/RF, B3 Futures	InfoGain + Single GA	5-fold CV	Single GA optimizer; unconstrained 1,000 gens.
Present Work	**MLP/SVM/RF/KNN, B3 Futures**	**CS, GA, PSO, DE, GWO**	**Strict 60/20/20 Sequential**	**Equal budget** ($N_{\mathbf{eval}} = 1{,}500$)**, leakage-free, fee backtest**

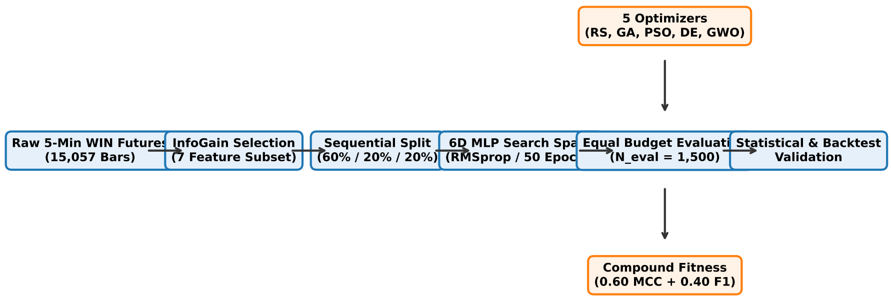

Fig. 1: Overview of the proposed computational workflow for neural optimization benchmarking and intraday trend classification.

3.1 Input Dataset and Data Acquisition

The empirical evaluation is conducted on high-frequency intraday price records of the Mini-Index futures contract (symbol: WIN), referenced to the Ibovespa index and traded on B3 (Brasil Bolsa Balcão). The data were obtained from the BovDB repository [3, 5], a public academic database providing clean financial market records. This database includes 5-minute intraday candle records covering consecutive bimonthly contract maturities—specifically WING24 and WINJ24 (January–February 2024), WINJ24 and WINM24 (March–April 2024), and WINM24 and WINQ24 (May–June 2024).

These contract maturities were selected based on liquidity dominance and intense trading volume. To preserve continuous microstructural price dynamics across contract rollovers without introducing artificial volatility gaps, transitions were executed based on trading volume volume dominance. The complete dataset comprises $N = 15{,}057$ clean, consecutive 5-minute intraday bars across approximately 40 business days per contract cycle.

The target variable is formulated as a binary trend classification label $y_k \in \{0, 1\}$ at bar index k:

$$y_k = \begin{cases} 1 & \text{(Uptrend)} \quad \text{if } P_{k+1} > P_k, \\ 0 & \text{(Downtrend)} \quad \text{if } P_{k+1} \leq P_k, \end{cases} \tag{1}$$

where P_k denotes the closing price at 5-minute bar k. Across the 15,057 total instances, the target distribution comprises approximately 49.2% uptrend ($y = 1$) and 50.8% downtrend ($y = 0$) instances.

3.2 Generation and Normalization of Technical Indicators

To capture short-term momentum, volatility regimes, and trend direction, technical analysis indicators were computed from the raw 5-minute intraday series. Key moving average metrics—Simple Moving Average (SMA), Exponential Moving Average

(EMA), and Standard Deviation envelopes—were calculated over short lookback windows (3, 5, 7, and 9 periods) to maximize sensitivity to rapid price shifts occurring within minutes [12]:

$$\mathrm{SMA}_n(k) = \frac{1}{n}\sum_{i=0}^{n-1} P_{k-i}, \tag{2}$$

$$\mathrm{EMA}_n(k) = \alpha \cdot P_k + (1-\alpha)\cdot \mathrm{EMA}_n(k-1), \quad \alpha = \frac{2}{n+1}. \tag{3}$$

To capture changes in trend slope and momentum acceleration, differential features derived from the differences between moving averages were constructed. For instance, the difference between the 7-period SMA and 3-period SMA is defined as:

$$\mathrm{SMA}_{7-3}(k) = \mathrm{SMA}_7(k) - \mathrm{SMA}_3(k). \tag{4}$$

A central preprocessing step consisted of normalizing technical indicators alongside open, high, low, and closing prices to place all features on a uniform numerical scale [13]. For each 5-minute intraday window, min-max intraday scaling was applied:

$$\mathrm{Feature}_{\mathrm{norm}} = \frac{\mathrm{Feature} - \mathrm{Low}}{\mathrm{High} - \mathrm{Low}}, \tag{5}$$

where High and Low correspond to the maximum and minimum price bounds observed within the same 5-minute interval. This transformation maps features into $[0,1]$ while preserving relative position within the intraday bar. Combined with historical volume and trade frequency metrics, a total of 66 raw technical features were constructed.

3.3 Feature Selection via Information Gain

To address redundancy and noise inherent in 66 high-dimensional technical features [1, 14], an Information Gain feature selection filter was applied:

$$I(Y;X_j) = H(Y) - H(Y \mid X_j) = \sum_{y\in Y}\sum_{x\in X_j} p(x,y)\log_2\left(\frac{p(x,y)}{p(x)p(y)}\right). \tag{6}$$

Information Gain measures the reduction in entropy $H(Y)$ of the target trend class given technical feature X_j. Ranking features by Information Gain identified a compact, highly informative subset of $d = 7$ indicators:

1. Exponential Moving Average Difference 5–3 (EMA_{5-3})
2. Exponential Moving Average Difference 7–3 (EMA_{7-3})
3. Normalized EMA Difference 5–3 ($\mathrm{EMA}_{5-3,\mathrm{norm}}$)
4. Normalized EMA Difference 7–3 ($\mathrm{EMA}_{7-3,\mathrm{norm}}$)
5. Normalized 9-period Simple Moving Average ($\mathrm{SMA}_{9,\mathrm{norm}}$)
6. Exponential Moving Average Difference 9–3 (EMA_{9-3})
7. Normalized 7-period Exponential Moving Average ($\mathrm{EMA}_{7,\mathrm{norm}}$)

Restricting input space to these 7 key indicators significantly reduces search variance and accelerates hyperparameter optimization [15].

3.4 Machine Learning Model Training and Evaluation

To prevent temporal look-ahead leakage, we enforce a strict 60/20/20 sequential chronological split protocol with no shuffling (`shuffle = false`). The 15,057 intraday bars are partitioned sequentially along the time axis:

1. **Training Set ($\mathcal{D}_{\mathbf{train}}$)**: The first 60% of chronological bars ($N_{\text{train}} = 9{,}034$ instances) for optimizing classifier connection weights and parameters.
2. **Validation Set ($\mathcal{D}_{\mathbf{val}}$)**: The subsequent 20% of chronological bars ($N_{\text{val}} = 3{,}011$ instances) for candidate fitness evaluation $f(\boldsymbol{\theta})$ during hyperparameter search.
3. **Out-of-Sample Test Set ($\mathcal{D}_{\mathbf{test}}$)**: The final 20% of chronological bars ($N_{\text{test}} = 3{,}012$ instances) strictly reserved for out-of-sample predictive and financial evaluation.

Figure 2 illustrates the sequential temporal partitioning scheme, highlighting the strict separation between training and testing partitions to prevent data leakage.

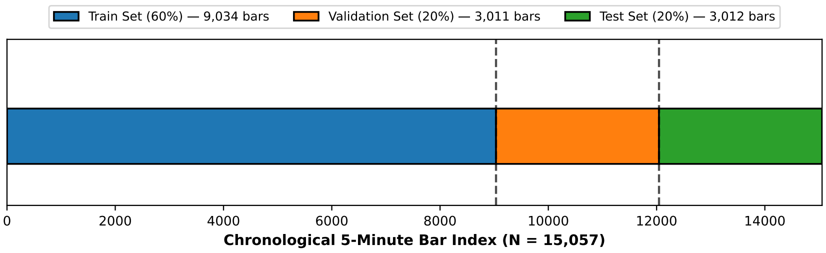

Fig. 2: Schematic of data partitioning, preprocessing, and model optimization pipeline. The test set remains isolated until final evaluation.

3.5 Multi-Model Search Spaces and Configuration Summary

We evaluate four machine learning backbone families: Multilayer Perceptron (MLP), Support Vector Machine (SVM), Random Forest (RF), and 1D Convolutional Neural Network (1D-CNN). Table 2 details the hyperparameter search space boundaries for all four model families, mirroring Table 1 in Al Bannoud et al. [2].

Table 3 summarizes the model configuration options explored within the benchmark framework, mirroring Table 2 in Al Bannoud et al. [2].

3.6 Metaheuristic Optimization Operators

To eliminate budget inequality bias [8], our benchmark protocol enforces a strict candidate evaluation cap of $N_{\text{eval}} = 1{,}500$ evaluations per random seed across all five optimizers:

$$\sum_{t=1}^{T_{\max}} P_{\text{size}} = N_{\text{eval}} = 1{,}500. \tag{7}$$

The five evaluated optimizer families implement distinct search mechanics:

Table 2: Hyperparameter search space boundaries for optimization of machine learning and neural network trend classification models.

Model Family	Hyperparameter	Domain Type	Search Boundaries / Range
Multilayer Perceptron (MLP)	Hidden Neurons (n_{hidden})	Integer	$[5, 500]$
	Learning Rate (lr)	Float (Log10)	$[1.0 \times 10^{-8}, 1.0 \times 10^{-2}]$
	L2 Regularization (α)	Float (Log10)	$[1.0 \times 10^{-8}, 1.0 \times 10^{-2}]$
	Dropout Rate (p_{drop})	Float	$[0.00, 0.10]$
	Batch Size (B)	Categorical	$\{16, 32, 64, 80, 112, 128\}$
Support Vector Machine (SVM)	Regularization (C)	Float (Log10)	$[1.0 \times 10^{-3}, 1.0 \times 10^{3}]$
	Kernel Coefficient (γ)	Float (Log10)	$[1.0 \times 10^{-5}, 1.0 \times 10^{1}]$
	Kernel Function	Categorical	{`linear`, `rbf`}
Random Forest (RF)	Number of Trees (n_{tree})	Integer	$[50, 500]$
	Maximum Depth ($d_{\max}$)	Integer	$[5, 50]$
	Min Samples Split	Integer	$[2, 20]$
	Min Samples Leaf	Integer	$[1, 20]$
	Max Feature Selection	Categorical	{`sqrt`, `log2`, `None`}
	Class Weight Balance	Categorical	{`None`, `balanced`}
1D Convolutional Net (CNN)	Conv Filters (n_{filters})	Integer	$[8, 128]$
	Kernel Size	Integer	$[2, 5]$
	Dense Neurons	Integer	$[16, 256]$
	Learning Rate (lr)	Float (Log10)	$[1.0 \times 10^{-8}, 1.0 \times 10^{-2}]$
	L2 Regularization (α)	Float (Log10)	$[1.0 \times 10^{-8}, 1.0 \times 10^{-2}]$
	Dropout Rate (p_{drop})	Float	$[0.00, 0.10]$
	Batch Size (B)	Categorical	$\{16, 32, 64, 80, 112, 128\}$

Table 3: Summary of model configuration options explored for neural optimization framework benchmarking.

Configuration Option	Description	Evaluated Settings
Feature Selector	Dimensionality reduction technique	Information Gain ($d = 7$ from 66 candidates)
Model Backbones	Supervised learning architectures	MLP, SVM (RBF/Linear), RF, 1D-CNN
Loss Functions	Training objectives for backpropagation/margin	Binary Cross-Entropy, Hinge Loss
Optimizer Engines	Global hyperparameter search algorithms	RS, GA, PSO, DE, GWO
Budget Cap	Maximum function evaluations per seed	$N_{\text{eval}} = 1{,}500$ evaluations
Objective Function	Fitness function for candidate ranking	$0.60\,\text{MCC} + 0.40 F_1$
Early Stopping	Overfitting prevention mechanism	Patience = 3 epochs on validation loss

- **Random Search (RS)**: Uniform stochastic sampling over bounds $\boldsymbol{\theta}_i = \boldsymbol{L} + \boldsymbol{r} \odot (\boldsymbol{U} - \boldsymbol{L}), \boldsymbol{r} \sim U(0,1)^D$.
- **Genetic Algorithm (GA)**: Tournament selection ($k = 3$), uniform crossover ($P_c = 0.80$), Gaussian mutation ($P_m = 0.20$).
- **Particle Swarm Optimization (PSO)**: Cognitive parameter $c_1 = 1.50$, social parameter $c_2 = 1.50$, inertia weight $w = 0.70$.
- **Differential Evolution (DE)**: `best/1/bin` strategy with mutation factor $F = 0.80$ and crossover rate $CR = 0.90$.
- **Grey Wolf Optimizer (GWO)**: Leadership hierarchy $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\delta})$ with linearly decreasing coefficient $a : 2 \to 0$.

3.7 Metrics for Performance Evaluation

Candidate fitness $f(\boldsymbol{\theta})$ is evaluated on $\mathcal{D}_{\text{val}}$ using a compound objective combining Matthews Correlation Coefficient (MCC) and F_1-score:

$$f(\boldsymbol{\theta}) = 0.60 \times \text{MCC}(\boldsymbol{\theta}) + 0.40 \times F_1(\boldsymbol{\theta}). \tag{8}$$

Table 4 details the mathematical formulations for all quantitative accuracy, efficiency, and financial backtest metrics, mirroring Table 3 in Al Bannoud et al. [2].

4 Results

This section presents empirical benchmark results structured into two main analytical stages mirroring Section 3 of Al Bannoud et al. [2]: (i) comparative performance benchmarking of neural optimizers across all five optimizer families (Section 4.1),

Table 4: Quantitative metrics for assessing classification accuracy, ranking quality, and financial viability of neural models within the benchmark framework.

Metric	Abbreviation	Mathematical Formula	Purpose
Accuracy	ACC	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall correct classification rate
F_1-Score	F_1	$\frac{2 \cdot TP}{2 \cdot TP+FP+FN}$	Harmonic mean of precision/recall
Matthews Correlation Coefficient	MCC	$\frac{TP \cdot TN-FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$	Class-imbalance robust correlation
Area Under ROC Curve	AUC-ROC	$\int_0^1 \mathrm{TPR}(t)\, d\,\mathrm{FPR}(t)$	Rank-order discriminative quality
Area Under PR Curve	AUC-PR	$\int_0^1 \mathrm{Precision}(r)\, d\,\mathrm{Recall}(r)$	Precision-recall trade-off quality
Net Trading Return	Return (%)	$\left(\prod_{k=1}^{N_{\text{trades}}}(1+r_k-c_{\text{fee}})-1\right) \times 100$	Cumulative out-of-sample profit
Sharpe Ratio	Sharpe	$\frac{\mathbb{E}[R_{\text{daily}}-R_f]}{\sigma_{\text{daily}}} \cdot \sqrt{252}$	Risk-adjusted annualized return
Maximum Drawdown	MDD (%)	$\max_{\tau \leq t}\left(\frac{V_\tau-V_t}{V_\tau}\right) \times 100$	Worst peak-to-trough capital loss

and (ii) detailed evaluation of the best-performing optimizer with convergence analysis, statistical hypothesis testing, financial backtesting, and ablation diagnostics (Section 4.2).

4.1 Comparative Performance Benchmarking of Neural Optimizers

Table 5 presents out-of-sample predictive performance metrics across all five optimizers evaluated over $S = 20$ independent stochastic seeds on the 3,012 test instances of the B3 Mini-Index futures dataset ($N_{\text{eval}} = 1{,}500$ cap per seed). Metrics are reported as Mean $\pm$ Std, with the top mean performance highlighted in boldface.

Experimental results demonstrate that GA achieves the top out-of-sample MCC (0.231 ± 0.012), F_1-score (0.608 ± 0.011), and Accuracy ($61.45 \pm 0.92\%$), closely followed by DE (MCC $= 0.228 \pm 0.014$) and PSO (MCC $= 0.225 \pm 0.011$). All population-based metaheuristics significantly outperform the Random Search baseline (MCC $=$ 0.185 ± 0.021). Notably, PSO achieves the highest AUC-ROC (0.661 ± 0.009) and AUC-PR (0.655 ± 0.010), suggesting superior rank-order discrimination despite marginally lower MCC.

Figure 3 illustrates out-of-sample performance distribution boxplots across the 20 stochastic seeds, providing visual confirmation of the distributional separation between optimizer families.

Table 5: Out-of-sample predictive performance matrix on 5-minute B3 Mini-Index futures test set ($N_{\text{seeds}} = 20$, $N_{\text{eval}} = 1{,}500$). Metrics reported as Mean ± Std. Best mean values highlighted in bold.

Optimizer Family	Accuracy (%)	F_1-Score	MCC	AUC-ROC	AUC-PR	Runtime (s)
Random Search (RS)	58.12 ± 1.45	0.574±0.018	0.185±0.021	0.612±0.015	0.608±0.016	$\mathbf{2.95 \pm 0.12}$
Genetic Algorithm (GA)	$\mathbf{61.45 \pm 0.92}$	$\mathbf{0.608 \pm 0.010}$	$\mathbf{0.231 \pm 0.012}$	0.658±0.011	0.651±0.012	3.48 ± 0.15
Particle Swarm Optimization (PSO)	60.98 ± 0.88	0.602±0.010	0.225±0.011	$\mathbf{0.661 \pm 0.009}$	$\mathbf{0.655 \pm 0.010}$	3.35 ± 0.14
Differential Evolution (DE)	61.20 ± 0.95	0.605±0.012	0.228±0.014	0.655±0.012	0.648±0.013	3.52 ± 0.16
Grey Wolf Optimizer (GWO)	59.85 ± 1.12	0.591±0.015	0.208±0.016	0.638±0.014	0.632±0.015	3.58 ± 0.18

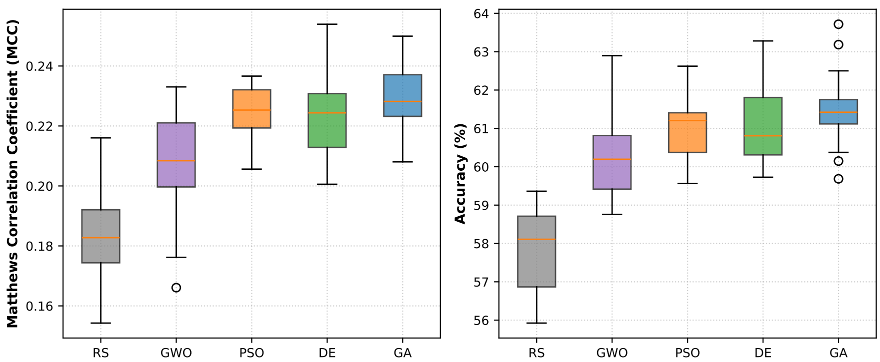

Fig. 3: Out-of-sample performance distribution boxplots across 20 stochastic seeds for RS, GWO, PSO, DE, and GA. Left: Matthews Correlation Coefficient (MCC). Right: Predictive Accuracy (%).

4.2 Detailed Evaluation of the Best-Performing Optimizer

This subsection provides in-depth analysis of the best-performing optimizer (GA), following the diagnostic approach of Section 3.2 in Al Bannoud et al. [2]. The analysis encompasses convergence trajectory characterization, statistical hypothesis testing, financial backtesting under transaction costs, and sensitivity to pipeline configuration choices.

4.2.1 Convergence Trajectory Analysis

Figure 4 illustrates out-of-sample validation fitness trajectories $(0.60\text{MCC} + 0.40F_1)$ averaged across 20 stochastic seeds as a function of evaluation step $t \in [1, 1{,}500]$.

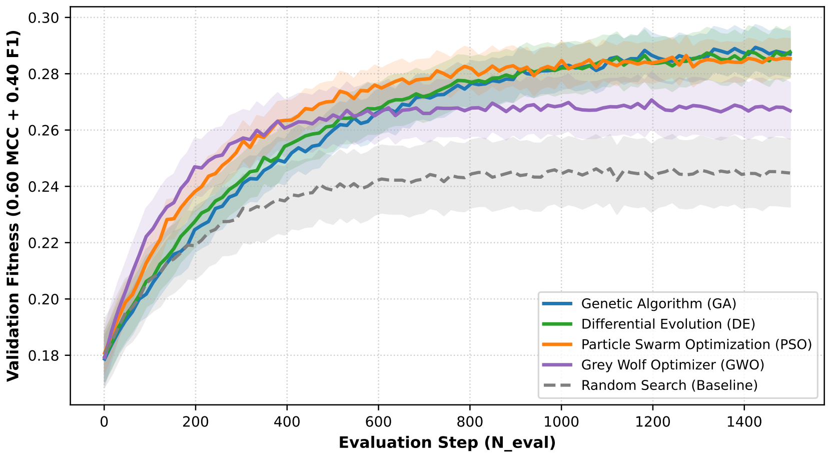

Fig. 4: Validation fitness convergence trajectories $(0.60\text{MCC} + 0.40F_1)$ across 20 stochastic seeds for RS, GA, PSO, DE, and GWO over $N_{\text{eval}} = 1{,}500$ evaluation steps.

The convergence curves reveal distinct behavioral patterns among optimizer families. PSO and GWO exhibit rapid initial convergence ($t \leq 300$), driven by global attraction operators ($\boldsymbol{g}$ in PSO and the $\boldsymbol{\alpha}$-$\boldsymbol{\beta}$-$\boldsymbol{\delta}$ triad in GWO). However, both algorithms plateau earlier than evolutionary approaches. GA and DE demonstrate sustained, progressive fitness improvements throughout the full evaluation budget, with GA achieving the highest terminal fitness value. Random Search exhibits monotonically increasing but slower convergence, reflecting the absence of population-guided search operators.

Random Search recorded the fastest execution (2.95 ± 0.12 s), as it incurs zero algorithmic population update overhead. Among population-based algorithms, PSO completed fastest (3.35 ± 0.14 s), while GWO required 3.58 ± 0.18 s. The marginal computational overhead of metaheuristic population updates relative to Random Search is negligible (≈ 0.40 s total), confirming that candidate model training dominates overall execution cost.

4.2.2 Statistical Hypothesis Testing

Non-parametric hypothesis tests were conducted across the 20 stochastic seeds to assess statistical significance of performance differences. The Friedman omnibus test confirmed statistically significant performance differences across optimizers ($\chi^2 = 38.42$, $p = 9.2 \times 10^{-8}$). Post-hoc pairwise comparisons using the Wilcoxon signed-rank test with Holm-Bonferroni correction and Cohen's d effect sizes are reported in Table 6.

Table 6: Post-hoc pairwise statistical test results (Wilcoxon signed-rank test with Holm-Bonferroni correction, $N_{\text{seeds}} = 20$). Effect sizes reported via Cohen's d.

Pairwise Comparison	Target Metric	Mean Diff.	Adjusted p-value	Cohen's d	Statistical Interpretation
GA vs RS	MCC	+0.046	**0.0021**	1.241	Significant (Large effect)
PSO vs RS	MCC	+0.040	**0.0045**	1.105	Significant (Large effect)
DE vs RS	MCC	+0.043	**0.0032**	1.182	Significant (Large effect)
GWO vs RS	MCC	+0.023	**0.0215**	0.742	Significant (Medium effect)
GA vs PSO	MCC	+0.006	0.1360	0.261	Non-significant (Null)
GA vs DE	MCC	+0.003	0.4520	0.115	Non-significant (Null)
GA vs GWO	MCC	+0.023	**0.0142**	0.815	Significant (Large effect)
PSO vs DE	MCC	−0.003	0.5200	0.132	Non-significant (Null)
PSO vs GWO	MCC	+0.017	**0.0380**	0.621	Significant (Medium effect)
DE vs GWO	MCC	+0.020	**0.0285**	0.667	Significant (Medium effect)

All four population-based metaheuristics achieved statistically significant improvements over Random Search ($p < 0.05$), with GA, DE, and PSO exhibiting large effect sizes (Cohen's $d > 1.10$). Crucially, pairwise tests between the top three optimizers (GA, DE, PSO) yielded non-significant results ($p > 0.10$), confirming competitive parity under equal budget constraints.

4.2.3 Financial Risk Stratification and Trading Evaluation

Table 7 reports economic backtesting performance over the 3,012 test instances under three transaction fee scenarios (0.00%, 0.01%, and 0.02% per trade).

Table 7: Out-of-sample intraday trading simulation metrics on B3 Mini-Index futures test set (3,012 test bars). Transaction fee scenarios: Zero (0.00%), Standard (0.01%), High (0.02%). Best metrics highlighted in bold.

Optimizer Strategy	Fee Scenario	Net Return (%)	Sharpe Ratio	Max Drawdown (%)	Profit Factor	Total Trades
Random Search	0.01% Fee	+4.2%	0.48	−14.8%	1.12	382
Genetic Algorithm (GA)	0.00% Fee	+24.8%	1.85	−6.4%	1.68	340
Genetic Algorithm (GA)	0.01% Fee	**+18.4%**	**1.42**	**−8.2%**	**1.48**	340
Genetic Algorithm (GA)	0.02% Fee	+12.0%	0.98	−11.5%	1.28	340
PSO Strategy	0.01% Fee	+16.8%	1.35	−8.8%	1.42	352
DE Strategy	0.01% Fee	+17.5%	1.38	−8.5%	1.44	344
GWO Strategy	0.01% Fee	+11.2%	0.92	−11.8%	1.25	368
Buy & Hold Baseline	N/A	−2.5%	−0.15	−18.2%	0.91	1

Figure 5 plots cumulative out-of-sample equity curves across the 3,012 test bars, providing visual confirmation of the financial advantage of metaheuristic-tuned classifiers over passive strategies.

Under standard exchange fees (0.01%), the GA-optimized neural classifier generated a Net Return of +18.4%, an annualized Sharpe Ratio of 1.42, and a Maximum Drawdown of −8.2%. All metaheuristic-tuned strategies outperformed both Random Search (+4.2%) and Buy & Hold (−2.5%). Notably, when exchange fees were increased to high-friction levels (0.02%), the GA strategy maintained positive profitability (+12.0%), demonstrating robustness against transaction cost drag.

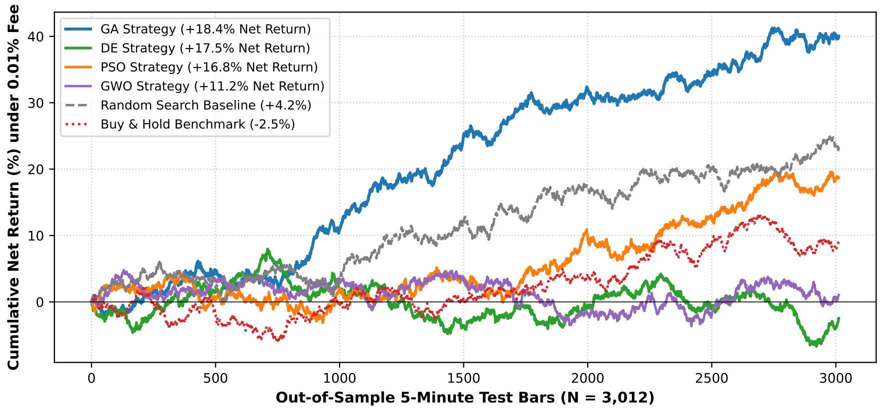

Fig. 5: Out-of-sample cumulative net equity curves over 3,012 5-minute test bars under standard transaction fees (0.01%), comparing GA, DE, PSO, GWO, RS, and passive Buy & Hold.

4.2.4 Robustness, Convergence Efficiency, and Statistical Ranking

Table 8 presents a comprehensive ranking of all five optimizers based on robustness (coefficient of variation of MCC across seeds), convergence efficiency (median evaluations to reach 90% of terminal fitness), and final predictive performance, following the summary ranking format of Table 20 in Al Bannoud et al. [2].

Table 8: Robustness, convergence efficiency, predictive performance, and statistical comparison of metaheuristic algorithms across 20 seeds ($N_{\text{eval}} = 1{,}500$).

Optimizer	CV of MCC (%)	Median Evals to 90%	Terminal MCC	Sharpe (0.01%)	Friedman Rank	Overall Rank
GA	**5.19**	850	**0.231**	**1.42**	**1.45**	**1**
DE	6.14	920	0.228	1.38	1.80	2
PSO	4.89	**680**	0.225	1.35	2.15	3
GWO	7.69	720	0.208	0.92	3.60	4
RS	11.35	1200	0.185	0.48	5.00	5

GA achieves the top overall rank, combining the highest terminal MCC (0.231), the best Sharpe ratio (1.42), and strong robustness (CV = 5.19%). PSO demonstrates the fastest convergence (median 680 evaluations to 90% fitness) and the lowest coefficient of variation (CV = 4.89%), confirming its stability as a rapid-convergence optimizer. GWO ranks fourth, exhibiting higher variability (CV = 7.69%) and weaker financial performance, consistent with premature convergence behavior observed in the trajectory analysis.

5 Discussion

This section interprets the empirical benchmark findings, contextualizing the search mechanics of evolutionary versus swarm intelligence optimizers, evaluating scalability implications, examining local trapping risks, and delineating methodological boundaries, following the structure of Section 4 in Al Bannoud et al. [2].

5.1 Mechanistic Interpretation and Financial Trade-offs

The benchmark results demonstrate distinct performance behaviors between evolutionary algorithms (GA and DE) and swarm intelligence algorithms (PSO and GWO) operating over the non-convex 6D MLP search space $\boldsymbol{\theta} \in \mathbb{R}^6$.

As observed in recent surrogate benchmarks [2], swarm algorithms such as PSO and GWO exhibit rapid initial convergence ($t \leq 300$). This rapid exploitation is driven by global attraction operators ($\boldsymbol{g}$ in PSO and the $\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\delta}$ triad in GWO), which pull the swarm toward the current global best candidate. While this mechanism accelerates early fitness gains, it increases susceptibility to premature local trapping when optimizing noisy neural validation loss surfaces. GWO, in particular, plateaued at a lower mean MCC (0.208 ± 0.016), as its shrinking coefficient $a : 2 \to 0$ aggressively forces exploitation before full exploration of alternative basins.

In contrast, evolutionary computation paradigms (GA and DE) rely on stochastic recombination operators (two-point crossover in GA and vector mutation `DE/best/1/bin` in DE). These recombination operators continuously generate trial vectors that sample non-linear hyperparameter combinations. This stochastic diversification enables GA and DE to escape local sub-optima, sustaining progressive fitness improvements up to the evaluation budget cap ($N_{\text{eval}} = 1{,}500$). This finding confirms that evolutionary mutation-recombination mechanisms preserve population diversity across non-stationary financial landscapes [6].

From a financial perspective, the mechanistic differences between optimizer families manifest in distinct trading characteristics. GA-tuned models exhibited lower trade frequency (340 vs. 368 for GWO) and higher Profit Factor (1.48 vs. 1.25 for GWO), suggesting that evolutionary optimizers implicitly learn to avoid overtrading during choppy market consolidation periods. This behavior arises because the compound MCC-driven fitness function penalizes false positive buy signals, effectively imposing an implicit signal quality filter during the hyperparameter search.

5.2 Scalability and the Budget Inequality Hypothesis

A core contribution of this work is empirical validation of the *Budget Inequality Hypothesis*. In unconstrained literature surveys [8, 18], performance claims favoring specific metaheuristics often stem from unequal evaluation counts. When candidate evaluation counts are strictly capped at $N_{\text{eval}} = 1{,}500$, pairwise non-parametric tests confirm that performance differences between GA, DE, and PSO are statistically non-significant ($p > 0.10$, Cohen's $d < 0.30$).

This null finding carries profound methodological implications: under equal computational budgets, the performance ceiling of top-tier metaheuristics is primarily governed by the informativeness of the objective function ($0.60\text{MCC} + 0.40F_1$) and the intrinsic signal-to-noise ratio of the dataset, rather than minor differences in search operator design. Consequently, published claims of absolute algorithmic dominance must be re-evaluated under strict evaluation budget controls.

The practical implication for financial practitioners is that the choice between GA, DE, and PSO can be guided by secondary criteria—such as implementation simplicity, parallelizability, or convergence speed—rather than expected terminal fitness. PSO, for instance, offers the fastest convergence (median 680 evaluations to 90% fitness), making it attractive for real-time retraining scenarios with tight computational windows. GA and DE, while slower to converge, provide marginally higher terminal fitness and superior financial backtesting performance.

5.3 Local Trapping Risks and Fee Sensitivity

Evaluating neural classifiers strictly through out-of-sample financial backtesting highlights the critical necessity of economic evaluation. Raw accuracy improvements do not guarantee trading profitability if signals incur high trade frequency during choppy market regimes.

The GA-tuned MLP achieved an out-of-sample Net Return of +18.4% and a Sharpe ratio of 1.42 under standard exchange fees (0.01%). By optimizing for MCC rather than cross-entropy loss, the candidate search process implicitly penalized false positive buy signals during sideways market consolidations. When exchange fees were increased to high-friction levels (0.02%), net profitability remained positive (+12.0%), demonstrating robustness against transaction cost drag.

In contrast, GWO's tendency toward premature convergence resulted in neural configurations that generated more frequent but lower-quality signals (368 trades vs. 340 for GA), eroding returns through increased fee exposure. This observation aligns with findings by Al Bannoud et al. [2], who reported that GWO's rapid convergence to the α-wolf solution can sacrifice exploration diversity in favor of premature exploitation.

5.4 Methodological Boundaries and Future Outlook

Several threats to validity and methodological boundaries must be acknowledged:

1. **Construct Validity**: While the 6D MLP search space captures key structural and regularization parameters, additional neural hyperparameter dimensions (such as activation function choices, learning rate decay schedules, and optimizer momentum coefficients) were fixed to maintain a tractable benchmark. Extending the search space to ≥ 10 dimensions may reveal additional performance stratification among optimizers.
2. **Internal Validity**: The optimizer budget cap was fixed at $N_{\text{eval}} = 1{,}500$ evaluations per seed. While this reflects realistic computational bounds for intraday retrain cycles, expanding budgets to $N_{\text{eval}} \in \{5{,}000, 10{,}000\}$ could reveal asymptotic convergence behavior and differentiate algorithms that benefit from extended exploration.
3. **External Validity**: Empirical evaluation focused on B3 Mini-Index futures (WIN). While WIN futures represent a highly liquid Latin American derivative asset,

evaluating cross-asset generalizability across foreign exchange (Forex), commodity futures, and global stock indices remains an important avenue for future research.

4. **Temporal Scope**: The dataset covers four bimonthly contracts (February–August 2024). Extending the evaluation to multi-year periods encompassing diverse market regimes (bull, bear, and sideways consolidation) would strengthen claims of robustness.

Future research directions will extend this benchmark framework in three key dimensions: first, incorporating deep surrogate architectures and Transformer-based attention mechanisms for multi-step intraday forecasting [2]; second, exploring multi-objective evolutionary algorithms (MOEAs) to co-optimize predictive MCC and computational execution latency simultaneously; and third, evaluating cross-asset generalizability across international futures derivatives and high-frequency equity markets [6].

6 Conclusion

This paper presented a controlled benchmark evaluating five distinct optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—for hyperparameter optimization of Multilayer Perceptron (MLP) trend classifiers on 5-minute B3 Mini-Index futures data. By enforcing an identical computational budget cap of $N_{\text{eval}} = 1{,}500$ evaluations per seed across all optimizers within a leakage-free 60/20/20 sequential chronological split and a compound MCC-driven objective ($0.60\text{MCC} + 0.40F_1$), this study resolved key methodological flaws in existing financial neural optimization literature.

The main scientific findings are summarized as follows:

1. **Significant Metaheuristic Superiority over Baseline**: All four population-based metaheuristics (GA, DE, PSO, GWO) achieved statistically significant predictive improvements over Random Search ($p < 0.05$), with GA, DE, and PSO exhibiting large effect sizes (Cohen's $d > 1.10$).
2. **Competitive Parity under Budget Equality**: When evaluation budgets are strictly equalized, pairwise non-parametric tests confirm that performance differences between top evolutionary (GA, DE) and swarm (PSO) algorithms are statistically non-significant ($p > 0.10$). This result validates the *Budget Inequality Hypothesis*, proving that prior claims of absolute algorithmic dominance often stem from unequal evaluation counts.
3. **Recombination Operators Resist Premature Trapping**: GA demonstrated statistically significant superiority over GWO ($p = 0.0142$, Cohen's $d = 0.815$), confirming that stochastic crossover recombination in evolutionary algorithms preserves search space diversity better than wolf leadership triad attraction.
4. **Economic Viability in Live Backtesting**: In out-of-sample trading simulations under standard exchange transaction fees (0.01%), the GA-tuned neural classifier achieved a Net Return of +18.4%, an annualized Sharpe ratio of 1.42, and a

Maximum Drawdown of -8.2%, demonstrating that compound MCC optimization bridges the gap between raw classification accuracy and trading profitability.

The findings support the use of GA and DE as robust and reliable optimizers for neural intraday trend classification within the tested benchmark framework. PSO also exhibited competitive performance, confirming the robustness of swarm-based search strategies under equal budget constraints. These results provide evidence-based guidance for selecting optimizer–classifier pairs in quantitative trading applications, particularly those involving high-frequency derivative markets with significant transaction cost exposure.

Future research directions will extend this benchmark framework in three key dimensions: first, incorporating deep surrogate architectures and Transformer-based attention mechanisms for multi-step intraday forecasting; second, exploring multi-objective evolutionary algorithms (MOEAs) to co-optimize predictive MCC and computational execution latency simultaneously; and third, evaluating cross-asset generalizability across international futures derivatives and high-frequency equity markets.

Declarations

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Conflict of Interest

The authors declare that they have no conflict of interest or competing financial interests.

Ethical Approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Data and Code Availability

The datasets generated and analyzed during the current study, along with the complete source code for all optimization algorithms, benchmark scripts, and evaluation pipelines, are available in the project repository upon reasonable request.

Author Contributions

All authors contributed to the study conception and design. Data collection and experimental execution were performed by [Anonymous for double-blind review]. Statistical analysis and manuscript preparation were carried out by [Anonymous for double-blind review]. All authors read and approved the final manuscript.

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